Less Reading, More Scoring!

ONE MARK QUESTIONS

OMQ

SECTION-A

Read in Minutes, Remember for Months!

TARGET MARKS:≥5

OMQ CHAPTERS

9 MARKS

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- Being Bullet Q Book, to make it short only 5 selective OMQ [5 MCQ, 5 FIB, 5 OWQ] are taken.

OMQ – Hints & Solutions

1. SOLUTIONS

I. MULTIPLE CHOICE Q'S (MCQ)

1. **Which of the following concentration term is independent of temperature?** [2]

   1. Molarity ✓**2. Molality**
   3. Mass-Volume percentage 4. Volume –Volume percentage

2. **Which of the following is an example of an ideal solution?** [3]

   1. $CS_2+CH_3COCH_3$ 2. $C_6H_5OH+C_6H_5NH_2$
   ✓**3.$C_6H_6+C_6H_5CH_3$** 4. $C_2H_5OH+CH_3COCH_3$

3. **The mole fraction of benzene in solution containing 30% by mass in CCl_4 is** [3]

   1. 0.5611 2. 0.9102 ✓**3. 0.4586** 4. 0.0214

4. **Which of the following solutions has the highest boiling point?** [2]

   1. 0.1 M KNO_3 ✓**2. 0.1 M Na_3PO_4** 3. 0.1 M $BaCl_2$ 4. 0.1 M K_2SO_4

5. **Which law explains the solubility of gases in liquids ?** [3]

   1. Boyle's law 2. Charles' law ✓**3.Henry's law** 4. Raoult's law

II. FILL IN THE BLANKS (FIB)

6. The units of boiling point elevation constant K_b are **K kg mol^{-1}** .
7. Solubility of a gas in liquid **increases** with increase in pressure.
8. For an ideal solution, $\Delta_{mix}H$ is equal to **Zero**.
9. Desalination plants work on the basis of the phenomenon called **reverse osmosis**.
10. Human blood is isotonic with **0.9% (w/v) NaCl**. solution (or) Saline. **[BMP]**

III. ONE WORD ANSWER Q'S (OWQ)

11. **What is the term used for a solution that does not follow Raoult's law?**

A: **Non-ideal solution** does not follow the Raoult's law.

---

12. **In which type of a solution, dynamic equilibrium exists between the dissolved solute and undissolved solute?**

A: In **saturated solution** dynamic equilibrium exists.

---

13. **What is the name given to binary liquid mixtures which have same composition in liquid and vapour phase and boil at a constant temperature?**

A: **Azeotropic mixtures** have same composition in liquid and vapour phase .

---

14. **In the expression $\Delta T_f = K_f m$, what is the name given to the constant K_f?**

A: K_f is called **Cryoscopic constant** (or) Molal depression constant.

---

15. **What is the name given to the solutions, when they have same osmotic pressure at a given temperature?**

A: **Isotonic solutions** have same osmotic pressure at a given temperature.

2. ELECTRO CHEMISTRY

I. MCQ

1. **During electrolytic refining of metals the refined metal is obtained at (on) the** [2]
   1) anode ✓**2)cathode** 3) surface of the electrolyte 4) Both 1 & 2

2. **The standard cell potential of a Daniel cell is** [4]
   1) 1.6 V 2) 1.9 V 3) 0.6 V ✓**4) 1.1V**

3. **In a dry cell which of the following is the electrolyte?** [1]
   ✓**1) $NH_4Cl + ZnCl_2$** 2) H_2SO_4+NaOH 3) NaOH+KOH 4) NaOH+HCl

4. **Which of the following is correct about cathode in an electro chemical cell?** [3]
   1) Oxidation occurs 2) Electrons loss occur ✓**3) Reduction occurs** 4) No reaction occurs

5. **A lead storage battery is mainly used in** [4]
   1) watches 2) automobiles 3) inverters ✓**4) Both 2 & 3**

II. FIB

6. 1 Faraday value is equal to **96487 (or) 96500** C mol^{-1}
7. Corrosion of iron is mainly due to **water** and **air**
8. The units of molar conductivity are **$S cm^2 mol^{-1}$ (or) $S\ m^2 mol^{-1}$**
9. Lead storage battery is an example of **secondary** cell. **[BMP]**
10. In Daniell cell the oxidation half reaction occurs at **anode**

III. OWQ

11. **Write an expression showing the relation between E°_{cell} and Gibbs energy change.**

A: $\Delta G^\circ = -nFE^\circ_{(cell)}$ is the relation between E°_{cell} and Gibbs energy change.

Here n is the no. of electrons involved in oxidation (or) reduction, E°_{cell} = standard EMF

---

12. **Give an example of an inert electrode.**

A: **Gold / Platinum / Graphite** are the examples of an inert electrode.

---

13. **What is the charge of an electron in coulombs?**

A: Charge of an electron in coulombs is **-1.602×10^{-19} coulombs**

---

14. **In a galvanic cell chemical energy is converted to electrical energy by which reaction?**

A: In a galvanic cell chemical energy is converted to electrical energy by **Spontaneous redox reaction.**

---

15. **What happens to the colour of a solution when a zinc wire is dipped in $CuSO_4$ solution?**

A: **Blue colour** of $CuSO_4$ **disappears** when a zinc wire is dipped in $CuSO_4$ solution.

3. CHEMICAL KINETICS

I. MCQ

1. **The unit of rate of a chemical reaction is generally expressed as:** [1]
   ✓**1.mol L^{-1} s^{-1}** 2. mol L^{-1} 3. mol 4. s^{-1}

2. **The rate law for a given reaction is determined from** [3]
   1. balanced chemical equation 2. thermodynamic data
   ✓**3.experimental data** 4. catalyst used

3. **The unit of rate constant of a first order reaction is** [1]
   ✓**1. s^{-1}** 2. L $mol^{-1}s^{-1}$ 3. mol L^{-1} s^{-1} 4. mol s^{-1}

4. **The half-life of a first-order reaction is given by** [3]
   1. $t_{1/2} = 1/k$ 2. $t_{1/2} = [A]_o/2k$ ✓**3. $t_{1/2} = 0.693/k$** 4. $t_{1/2} = k/0.693$

5. **Catalyst increases the rate of a reaction by** [1]
   ✓**1. decreasing activation energy** 2. increasing temperature
   3. increasing activation energy 4. increasing concentration

II. FIB

6. The sum of powers of the concentration terms in the rate law expression is called **order** of that chemical reaction.
7. The overall rate of the reaction is controlled by the slowest step of the reaction and is called the **rate determining step**
8. The number of collisions per second per unit volume of the reaction mixture is known as **collision frequency** **[BMP]**
9. $k = A\, e^{-Ea/RT}$ where A is called **Arrhenius factor or frequency factor**
10. For a first order reaction the following graph is obtained.
    From this graph slope is equal to **k/2.303**

III. OWQ

11. **What is the molecularity of the elementary reaction: A + B → Products?**

A: Molecularity of the given elementary reaction is **2 (two)**

---

12. **Which reactions takes place in one step?**

A: **Elementary reactions** takes place in one step.

---

13. **In a zero - order reaction, if initial concentration of reactant is 0.4 mol/L and rate constant is 0.1 mol $L^{-1}s^{-1}$, what is its $t_{1/2}$?**

A: Given that $[A_0]$=0.4 m/L, K=0.1 m.L^{-1} s^{-1} · For Zero order reaction $t_{1/2} = \frac{[A_0]}{2K} = \frac{0.4}{2\times0.1} = 2$ s

---

14. **What is the unit of rate constant for a zero-order reaction?**

A: Unit of rate constant for a zero-order reaction is **mol L^{-1} s^{-1}**

---

15. **Collision theory is based on which theory of gases?**

A: Collision theory is based on **Kinetic molecular theory of gases.**

4. THE d- AND f- BLOCK ELEMENTS

I. MCQ

1. The correct electronic configuration of Cr is [2]
   1. $[Ar]3d^4\ 4s^2$ **2. $[Ar]3d^5\ 4s^1$** 3. $[Ar]3d^6\ 4s^o$ 4. $[Ar]3d^3\ 4s^2$
2. Which of the following is not regarded as a transition element? [4]
   1. Cr 2. Mn 3. Fe **4. Zn**
3. The common oxidation state of d-block elements is [2]
   1. +3 **2. +2** 3. +1 4. +4
4. Which of the following set of elements represent coinage metals? [1]
   1. Cu, Ag, Au 2. Cu, Fe, Zn 3. Ag, Fe, Cr 4. Au, Fe, Zn
5. Transition metals form large number of complex compounds. This is due to [4]
   1. small size 2. high nuclear charge
   3. presence of vacant d-orbitals **4. All the above**

II. FIB

6. The atomic radii of 5d series of elements is same as that of **4d series** series of elements.
7. In aqueous solution the colour of the Cu^{2+} ion is **blue**
8. Ce^{4+} is a strong **oxidizing** agent.
9. Most of the radioactive elements are present in **5f series / f-block** series/ block of periodic table.
10. The common oxidation state of lanthanides is **+3**

III. OWQ

11. Name the transition element which does not exhibit variable oxidation states?

A: **Sc (Scandium)**does not exhibit variable oxidation states.

---

12. Which substances are attracted very strongly by applied magnetic field?

A: **Ferro Magnetic substances** are attracted **very strongly**.

---

13. What is the other name of f-block elements?

A: **Inner transition elements** is the other name of **f-block elements.**

---

14. What is the catalyst used in the manufacture of high-density polythene?

A: **Zeigler - Natta Catalyst** is used in the manufacture of **high-density polythene.**

---

15. Which metals are involved in the formation of an alloy called Brass?

A: **Cu and Zn** are used in the formation of **Brass.**

5. CO-ORDINATION COMPOUNDS

I. MCQ

1. **Which of the following coordination compound contains cobalt metal?[BMP]** [3]

   1. Chlorophyll 2. Hemoglobin ✓**3. Vitamin B_{12}** 4. Myoglobin

2. **Which of the following is not a double salt?** [3]

   1. $FeSO_4.(NH_4)_2SO_4.6H_2O$ 2. $KCl.MgCl_2 6H_2O$

   ✓**3. $PtCl_4.2HCl$** 4. $KAl(SO_4)_2.12H_2O$

3. **Lewis acid in the complex ion $[CoCl(NH_3)_5]^{2+}$ is** [4]

   1. Cl^- 2. NH_3 3. Co^{2+} ✓**4. Co^{3+}**

4. **Which ligand gives a stable complex?** [4]

   1. SCN^- 2. Cl^- 3. H_2O ✓ **4. Ethylene diamine(en)**

5. **Which complex shows geometrical, optical and also structural isomerism?** [2]

   1. $[Co(en)_3]Cl_3$ ✓**2. $[CoCl_2(en)_2]Br$** 3. $[Co(en)(NH_3)_2Cl_2]Cl$ 4. $[PdCl_2(NH_3)_2$

II. FIB

6. According to Werner's theory of coordination compounds primary valences of metal ion are satisfied by **negative ions**

7. Mohr's salt formula is **$FeSO_4(NH_4)_2SO_4.6H_2O$**

8. The coordination sphere in potassium ferrocyanide complex is **$[Fe(CN)_6]^{4-}$**

9. The IUPAC name of $K_2[PdCl_4]$ is **Potassium tetrachloridopalladate (II)**

10. M-CO bond in metal carbonyls is very strong due to **Synergic** effect. (or) π-back bonding.

III. OWQ

11. **Which platinum complex can effectively inhibit the growth of tumors?**

A: **Cis-platin** can effectively inhibit the growth of tumors. **Ex:** Cis $[Pt(NH_3)_2Cl_2]$

---

12. **Facial and Meridional isomers come under which category of isomerism?**

A: Facial and Meridional isomers show **Geometrical isomerism.**

---

13. **What is the geometry of $Fe(CO)_5$ complex?**

A: Geometry of $Fe(CO)_5$ is **Trigonal bipyramidal.**

---

14. **Which complex ion is formed when undecomposed AgBr of the film dissolves in hypo solution.**

A: **$[Ag(S_2O_3)_2]^{3-}$** is formed when AgBr film dissolves in hypo solution.

---

15. **How many unpaired electrons are there in $[NiCl_4]^{2-}$ ion?**

A: **Two unpaired electrons** are present in $[NiCl_4]^{2-}$ ion.

Ni^{2+} : $[Ar]4s^0 3d^8$

6. HALOALKANES & HALOARENES

I. MCQ

1. What is the general formula for haloalkanes? (X=halogen atom, n=1, 2, 3...) [2]
   1) $C_nH_{2n}X$ ✓2) $C_nH_{2n+1}X$ 3) $C_nH_{2n-1}X$ 4) $C_nH_{2n-3}X$
2. A primary alkyl halide would prefer to undergo [2]
   1) S_N1 reaction ✓2) S_N2 reaction 3) α–elimination 4) racemization
3. Which of the following is the suitable medium for preparing Grignard reagents? **[BMP]** [2]
   1) Dry acetone ✓2) Dry ether 3) Concentrated HCl 4) Alcoholic KOH
4. Which of the following alkyl halide will undergo S_N1 reaction most readily? [4]
   1) $(CH_3)_3C—F$ 2) $(CH_3)_3C—Cl$ 3) $(CH_3)_3C—Br$ ✓4) $(CH_3)_3C—I$
5. Chlorobenzene is [1]
   ✓1) less reactive than benzyl chloride 2) more reactive than ethyl bromide
   3) nearly as reactive as methyl chloride 4) more reactive than isopropyl chloride

II. FIB

6. Our body produces an iodine containing hormone, thyroxine, the deficiency of which causes a disease called **Goiter**
7. Alkyl iodides are often prepared by the reaction of alkyl chlorides/ bromides with NaI in dry acetone. This reaction is known as **Finkelstein** reaction.
8. Groups like cyanides and nitrites possess two nucleophilic centers and are called **ambident nucleophiles**
9. The compounds which rotate the plane polarized light are called **Optically active or Chiral**
10. The compounds which are obtained by the reaction of haloalkanes with magnesium metal in dry ether are called **Grignard Reagents**

III. OWQ

11. Preservation of the spatial arrangement of bonds to an asymmetric center during a chemical reaction or transformation is called

A: Preservation of the spatial arrangement of bonds is called **Retention of Configuration.**

12. Diazonium salt with cuprous chloride or cuprous bromide results in the replacement of the diazonium group by –Cl or –Br. This reaction is called

A: **Sandmeyer reaction** is the replacement of the diazonium group by –Cl or –Br.

13. Synthesis of alkyl fluorides by heating an alkyl chloride/bromide in the presence of a metallic fluoride such as AgF is named as

A: **Swarts reaction** is the synthesis of alkyl fluorides by heating an alkyl chloride/bromide.

14. If the compound rotates the plane of plane polarized light to the right, i.e., clockwise direction, it is called

A: **Dextro rotatory** compound rotates the plane polarized light to the right.

15. The stereoisomers which are related to each other as non- superimposable mirror images are called

A: **Enantiomers** are non- superimposable mirror images

7.ALCOHOLS, PHENOLS & ETHERS

I. MCQ

1. **Glycerol is a: [BMP]** [4]
   1) dihydric alcohol 2) monohydric alcohol 3) hexahydric alcohol ✓**4) trihydric alcohol**
2. **Which of the following can work as a dehydrating agent for alcohols?** [4]
   1) Conc. H_2SO_4 2) Anhydrous Al_2O_3 3) H_3PO_4 ✓**4) All of these**
3. **Which of the following shows strong hydrogen bonding?** [3]
   1) Ethyl amine 2) Ammonia ✓**3)Ethyl alcohol** 4) Diethyl ether
4. **When phenol is treated with excess bromine water it gives** [4]
   1) m-bromophenol 2) o- and p-bromophenol
   3) 2,4-dibromophenol ✓**4) 2,4,6-tribromophenol**
5. **Which of the following is formed when phenol is exposed to air?** [1]
   ✓**1) Benzoquinone** 2) Benzyl quinone 3) Acetophenone 4) Benzene

II. FIB

6. Isopropyl benzene is commonly called **cumene.**
7. Boiling points of alcohols and phenols are higher in comparison to other classes of compounds due to **intermolecular hydrogen bonding**
8. Alcohols and phenols react with carboxylic acids to form **esters**
9. Lucas reagent is **Conc. HCl/$ZnCl_2$** (anhydrous)
10. On treating phenol with chloroform in the presence of sodium hydroxide, salicylaldehyde is formed. This reaction is known as **Reimer-Tiemann reaction**

III. OWQ

11. **Phenol on heating with zinc dust gives a compound X. What is X?**

A: Phenol on heating with zinc dust gives **Benzene (C_6H_6)**

---

12. **When an alkyl halide is allowed to react with sodium alkoxide, ether is formed. What is the name given to this reaction?**

A: **Williamson synthesis:** Alkyl halide + sodium alkoxide→ Ether
(R'–X + R–O–Na → R–O–R' + NaX)

---

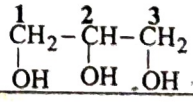
13. **In the preparation of phenol from cumene, what is the byproduct obtained?**

A: The byproduct obtained in the preparation of phenol from cumene is **Acetone (CH_3COCH_3)**

---

14. **What is the IUPAC name of glycerol?**

A: IUPAC name of glycerol is **Propane-1,2,3-triol.**



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15. **Which class of compounds are obtained when alkenes react with water in the presence of acid as catalyst?**

A: Alkenes when react with water in the presence of acid as catalyst gives **Alcohols.**

$$CH_2=CH_2+H_2O \xrightarrow{H^+} CH_3CH_2OH$$

8. ALDEHYDES, KETONES & CABOXYLIC ACIDS

I. MCQ

1. With which of the following reagents both ethanal and propanone react? [4]
   1) Tollens' 2) Schiff's 3) Fehling ✓**4) Grignard**

2. Cannizzaro reaction is not given by [1]
   ✓**1) CH_3CHO** 2) PhCHO 3) HCHO 4) $(CH_3)_3C–CHO$

3. Which of the following is a dicarboxylic acid? [1]
   ✓**1) oxalic acid** 2) benzoic acid 3) salicylic acid 4) picric acid

4. Common reagent used to detect aldehydes and ketones is [2]
   1) Lucas reagent ✓**2) 2, 4-DNP** 3) Baeyer's reagent 4) Tollens' reagent

5. Choose the weakest acid among the following [4]
   1) FCH_2COOH 2) Cl_2CCOOH 3) CH_3COOH ✓**4) CH_3CH_2COOH**

II. FIB

6. IUPAC name of formaldehyde is **methanal (HCHO)**
7. Chromium trioxide (CrO_3) in acidic (H_2SO_4) media is called **Jones** reagent
8. Tollens' reagent is **freshly prepared ammonical silver nitrate solution**
9. Carboxylic acids having an α-hydrogen are halogenated at the α-position on treatment with chlorine or bromine in presence of small amount of phosphorus to give α- halocarboxylic acids. This reaction is known as **Hell-Volhard-Zelinsky reaction (HVZ reaction)**
10. IUPAC name of Mesityl oxide is **4-methylpent-3-en-2-one**

III. OWQ

11. What is name of the product formed when benzene is reacted with CO and HCl in the presence of anhydrous $AlCl_3$?
    A: **Benzaldehyde** is obtained when benzene is reacted with CO and HCl.

---

12. What is Rochelle salt? **[BMP]**
    A: **Rochelle salt:** Sodium potassium tartarate +Sodiumhydroxide
    (A double salt of Tartaric acid)

---

13. Which acid is used as solvent in food industry as vinegar?
    A: Vinegar: **Ethanoic acid (CH_3COOH or acetic acid).**

---

14. Aldehydes and ketones undergo what type of addition reactions?
    A: Aldehydes and ketones undergo **Nucleophilic addition reactions.**

---

15. Phthalic acid on reaction with ammonia followed by strong heating gives a compound 'X' along with elimination of NH_3. What is the name of that compound 'X'?
    A: **phthalimide**

9. AMINES

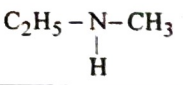
1. **Which of the following amines cannot be prepared by Gabriel phthalimide reaction?**
   1) Benzylamine ✓**2) Aniline** 3) Ethylamine 4) Methylamine [2]
2. **Classification of amines into 1°, 2°, 3° is based on the** [3]
   1) number of amino groups 2) nature of carbon atom
   ✓**3) degree of substitution at nitrogen** 4) degree of unsaturation
3. **The number of primary amines possible for a compound with molecular formula $C_4H_{11}N$ are**
   1) 1 2) 2 3) 3 ✓**4) 4** [4]
4. **Which among the following has the lowest pK_b value in aqueous medium?** [3]
   1) NH_3 2) CH_3NH_2 ✓**3) $(CH_3)_2NH$** 4) $(CH_3)_3N$
5. **Aniline reacts with the following to give benzanilide** [3]
   1) Benzyl chloride 2) Benzal chloride ✓**3) Benzoyl chloride** 4) Benzo tri chloride

II. FIB

6. The shape of trimethyl amine is **pyramidal**
7. Benzenesulphonyl chloride ($C_6H_5SO_2Cl$), is also known as **Hinsberg's reagent**
8. Carbylamine reaction or isocyanide test is used for the identification of **primary amines or 1° amines**
9. The conversion of primary aromatic amines into diazonium salts is known as **diazotisation**
10. Coupling reaction of aryl diazonium salts with phenols or aryl amines gives rise to the formation of **azodyes**

III. OWQ

11. **What is the IUPAC name of ethyl methyl amine?** **[BMP]**



A: IUPAC name of ethyl methyl amine is **N–methyl ethanamine.**

---

12. **The process of cleavage of the C–X bond of alkyl halides by ammonia molecule is known as**

A: **Ammonolysis** is the cleavage of the C–X bond of alkyl halides by ammonia molecule.

---

13. **Aniline and other arylamines are usually colourless but get coloured on storage why?**

A: Aniline and other arylamines get coloured on storage due to **Atmospheric oxidation.**

---

14. **Tertiary amines like trimethylamine are used as**

A: Tertiary amines are used to **attract flies or insects.**

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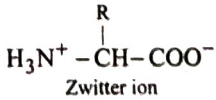
15. **A molecule that has both positive and negative regions of charge is called**

A: **Zwitter ion** is the molecule that has both positive and negative regions of charge.



Zwitter ion

10. BIOMOLECULES

I. MCQ

1. **Which of the following is not an example of a hormone?** [3]
   1) Insulin 2) Estrogen ✓**3) Pyridoxine** 4) Androgen
2. **Which of the following produces only glucose units on complete hydrolysis?** [2]
   1) Lactose ✓**2) Maltose** 3) Raffinose 4) Sucrose
3. **Which compound inverts its optical rotation on hydrolysis?** [2]
   1) Glucose ✓**2) Sucrose** 3) Maltose 4) Fructose
4. **Which amino acid is basic in nature?** [2]
   1) Glutamic acid ✓**2) Lysine** 3) Tyrosine 4) Leucine
5. **Which vitamin deficiency causes convulsions?** [4]
   1) Vitamin C 2) Vitamin D 3) Vitamin E ✓**4) Vitamin B_6**

II. FIB

6. In sucrose molecule, glucose and fructose are connected between **C_1 and C_2** atoms respectively.
7. Curdling of milk is an example of **denaturation of protein.**
8. Complementary base for the adenine present in DNA is **thymine.**
9. Rickets disease is caused by the deficiency of vitamin **D** .
10. Hormones are molecules that act as **intercellular messengers.**

III. OWQ

11. **Which linkage connects two monosaccharide units?** **[BMP]**

A: **Glycosidic linkage** connects two monosaccharide units.

---

12. **Which non-essential amino acid is optically inactive?**

A: **Glycine** is optically inactive non-essential amino acid .

---

13. **What type of bonding helps in stabilizing the α-helix structure of proteins?**

A: **Hydrogen bonding** helps in stabilizing the α-helix structure of proteins

---

14. **Name the disease that occurs due to deficiency of ascorbic acid.**

A: **Scurvy (bleeding gums)** occurs due to deficiency of ascorbic acid.

---

15. **Increased level of which hormone causes hyperthyroidism ?**

A: Increased level of **Thyroxine** causes hyperthyroidism.

SHINING STARS

STAR QUESTIONS PLUS

SOLUTIONS

VSAQ

175. What is van't Hoff's factor?

How is it related to 'α' in the case of a binary electrolyte (1:1)?

A: **Van't Hoff's factor(i)**: It is the ratio of the experimental value of the colligative property to the calculated value of the colligative property.

Incase of a binary electrolyte (1:1), the relation between α and i is as follows:

$i = \frac{1+(n-1)\alpha}{1}$. Degree of dissociation, $\alpha = \frac{i-1}{n-1}$

The Van't Hoff factor i accounts for dissociation or association of solute.

SAQ

176. If the osmotic pressure of glucose solution is 1.52 bar at 300K. What would be its concentration if R=0.083 L bar mol^{-1} K^{-1} ?

A: Given osmotic pressure Π=1.52, R = 0.083 L bar mol^{-1} K^{-1}, T= 300K, Concentration C = ?

Formula: $\Pi = CRT \Rightarrow C = \frac{\Pi}{RT} = \frac{1.52}{0.083 \times 300} = 0.061\,M$

---

177. How is molar mass related to the depression in freezing point of a solution?

A: Depression in Freezing point $\Delta T_f = K_f \times m$, where m= molality of the solution

$$\Rightarrow \Delta T_f = K_f \times \frac{w_s}{M} \times \frac{1000}{w_o(g)} \Rightarrow M_s = \frac{K_f \times w_s \times 1000}{\Delta T_f \times w_o}$$

Here, M_s = molar mass of solute, K_f = molal depression constant

w_s = Mass of solute in grams, ΔT_f = depression in freezing point, w_o = Mass of solvent in grams

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ELECTRO CHEMISTRY (SAQ)

178. On what factors the electrical conductance of an aqueous solution of electrolyte depends?

A: The conductance of the solution of an electrolyte depends on the following factors

i) **Nature of the electrolyte:** Strong electrolytes ionize almost completely in the solution and hence conduct electricity to a large extent. Whereas weak electrolytes ionise to a small extent and hence conduct electricity to a small extent.

ii) **Size of the ions :** Greater the size of the ions or greater the solvation of the ions, lesser is the conductance.

iii) **Nature of the solvent and its viscosity:** Electrolytes ionise more in a polar solvent. Hence, greater the polarity of the solvent, greater is the ionization and hence greater is the electrical conductance. Also greater the viscosity of a solvent, lesser is the conductance.

iv) **Concentration of the solution:** Higher the concentration of the solution, lesser is the conductance. This is because in a weak electrolyte, the ionization is less whereas in a strong electrolyte, the interionic attractions are higher at higher concentration.
With dilution, incase of weak electrolytes ionization increases and incase of strong electrolytes, interionic attraction decreases and hence conductance increases.

v) **Temperature:** On increasing the temperature, the dissociation of electrolyte increases and hence the conductance increases.

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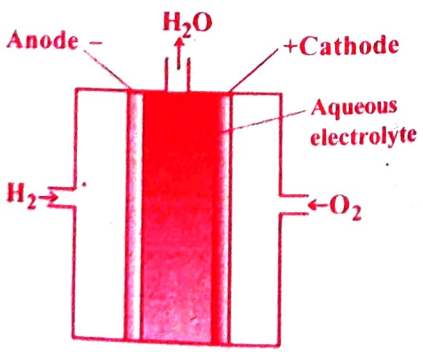
179. What are fuel cells? How are they different from galvanic cells?
Give the construction of H_2-O_2 fuel cell.

A: **A fuel cell** is a device that converts the energy from combustion of fuel into electrical energy. The fuel cell like any other electrochemical cell has two electrodes and an electrolyte.
Oxidation of fuel occurs at the anode: Fuel → oxidation product + ne^-
The oxidant gets reduced at the cathode: Oxidation + ne^- → Reduction product.
Fuel cells are different from conventional galvanic cells in the following aspects:

a) In a fuel cell, reactants are fed into the cell and products are removed from the cell. These do not form integral part of cell as in a conventional galvanic cell.

b) Chemical energy is not stored in a fuel cell, whereas in a galvanic cell chemical energy is converted into electrical energy.

H_2–O_2 fuel cell: This fuel cell consists of porous carbon electrodes suspended in concentrated NaOH solution.



Catalysts like finely divided platinum or palladium is incorporated into the electrodes for increasing the rate of electrode reactions.

H_2 and O_2 gases are bubbled at the surface of the electrodes.

The electrode reactions:

At cathode: $O_{2(g)} + 2H_2O + 4e^- \rightarrow 4OH^-_{aq}$

At anode: $2H_{2(g)} + 4OH^-_{aq} \rightarrow 4H_2O_{(\ell)} + 4e^-$

Overall reaction: $2H_{2(g)} + O_{2(g)} \rightarrow 2H_2O_{(\ell)}$

180. Explain the variation of molar conductivity with the change in the concentration of the electrolyte. Give reasons.

A: Molar conductivity of electrolytes generally, increases with dilution.

(i) For Strong Electrolytes (NaCl, KCl):

Λ^o_m increases **slightly** with dilution.

The increase is **gradual and linear**

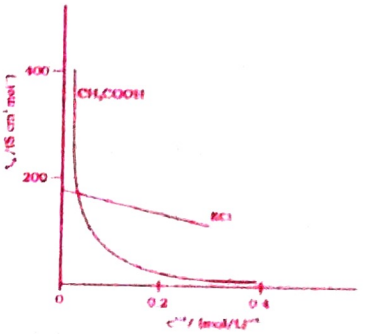
Reason:

Strong electrolytes are **almost completely dissociated** even at higher concentrations

On dilution:

Inter-ionic attractions decrease, Ion mobility increases

Hence, conductivity increases slightly



(ii) For Weak Electrolytes (CH_3COOH, NH_4OH):

Λ^o_m increases **sharply** with dilution

Reason: Weak electrolytes are **partially ionised**

On dilution: Degree of ionisation increases (Le Chatelier's principle)

More ions are produced. Hence, conductivity increases rapidly.

Graphical representation of the variation of Λ_m vs $\sqrt{C}$

For a strong electrolyte (KCl) and a weak electrolyte (CH_3COOH) is shown in the graph.

Strong electrolyte - straight line (small increase)

Weak electrolyte - curved line (steep increase)

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181. Explain with a suitable example the relation between the Gibbs energy of chemical reaction (G) and the functioning of the electrochemical cell.

A: Electrical work done in one second in an electrochemical cell is equal to decrease in its Gibbs energy. Electrical work done = decrease in free energy.

In the case of an electrochemical cell, the electrical work done is equal to the electrical energy produced. This in turn is equal to the product of the quantity of electricity flowing in the circuit and the EMF of the cell.

Now, for every one mole of electrons transferred in the cell reaction, the quantity of electricity that flows through the cell is one faraday (1F = 96,500 Coulomb).

Hence, if n moles of electrons are transferred, in any cell reaction, the quantity of electricity flowing = nF Faraday

If E_{cell} represents the EMF of the cell, then electrical work done = nF E_{cell}

Hence $-\Delta_r G = nF\ E_{cell}$

For comparing different cells, standard cell potentials are used which are represented by E^0_{cell}

The corresponding free energy change is called the standard change of the reaction, represented by $\Delta_r G^0$.

Hence we can write $-\Delta G^0 = nF\ E^0_{cell}$

Thus for a cell reaction $Zn_{(s)} + Cu^{+2}_{(aq)} \rightarrow Zn^{+2}_{(aq)} + Cu_{(s)}$

$\Delta_r G^0 = -2F\ E^0_{cell}$ [$\because$ n=2 = no.of electrons involved]

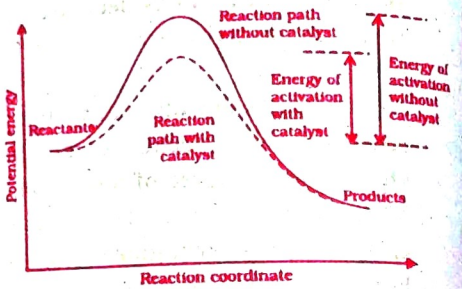
CHEMICAL KINETICS (SAQ)

182. Discuss the effect of catalyst on the kinetics of a chemical reaction with a suitable diagram.

A: A catalyst is a substance which increases the rate of a reaction without itself undergoing any permanent chemical change.

Ex: MnO_2 acts as catalyst in the following reaction to increase its rate .

$$2KClO_3 \xrightarrow{MnO_2} 2KCl + 3O_2$$



The action of the catalyst can be explained by intermediate complex theory. According to this theory, a catalyst participates in a chemical reaction by forming temporary bonds with the reactants . It results in the formation of an intermediate complex. This has a transitory existence and decomposes to yield products and the catalyst.

Catalyst takes the reaction in a new path having low activation energy. When the activation energy decreases, the rate of reaction increases.

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183. Define average rate of a reaction. How is the rate of reaction expressed in terms of change in the concentration of reactants and products for the following reactions.

1) $2HI_{(g)} \rightarrow H_{2(g)} + I_{2(g)}$ 2) $Hg_{(l)} + Cl_{2(g)} \rightarrow HgCl_{2(s)}$

3) $5Br^-_{(aq)} + BrO^-_{3(aq)} + 6H^+_{(aq)} \rightarrow 3Br_{2(aq)} + 3H_2O_{(l)}$

A: **Average rate of a reaction** is defined as the rate of change of concentration per unit time.

$$\text{Average time} = \frac{\text{Change in concentration in give time}}{\text{Time taken}} = \frac{\Delta x}{\Delta t} \text{ (or) } -\frac{\Delta[R]}{\Delta t} = +\frac{\Delta[P]}{\Delta t}$$

1) $2HI_{(g)} \rightarrow H_{2(g)} + I_{2(g)}$ Average rate $= -\frac{1}{2}\frac{\Delta[HI]}{\Delta t} = +\frac{\Delta[H_2]}{\Delta t} = +\frac{\Delta[I_2]}{\Delta t}$

2) $Hg_{(l)} + Cl_{2(g)} \rightarrow HgCl_{2(s)}$ Average rate $= -\frac{\Delta[Hg]}{\Delta t} = -\frac{\Delta[Cl_2]}{\Delta t} = +\frac{\Delta[HgCl_2]}{\Delta t}$

3) $5Br^-_{(aq)} + BrO^-_{3(aq)} + 6H^+_{(aq)} \rightarrow 3Br_{2(aq)} + 3H_2O_{(l)}$

$$\text{Average rate} = -\frac{1}{5}\frac{\Delta[Br^-]}{\Delta t} = -\frac{\Delta[BrO_3^-]}{\Delta t} = -\frac{1}{6}\frac{\Delta[H^+]}{\Delta t} = +\frac{1}{3}\frac{\Delta[Br_2]}{\Delta t} = +\frac{1}{3}\frac{\Delta[H_2O]}{\Delta t}$$

184. What is rate equation? How is it obtained? Write the rate equations for

1) $2NO_{(g)} + O_{2(g)} \rightarrow 2NO_{2(g)}$ 2) $CHCl_3 + Cl_2 \rightarrow CCl_4 + HCl$

3) $CH_3COOC_2H_{5(l)} + H_2O_{(l)} \rightarrow CH_3COOH_{(aq)} + C_2H_5OH_{(aq)}$

A: **Rate equation (rate law)** is the expression in which reaction rate is given in terms of molar concentration of reactants with each term raised to some power, which may or may not be same as the stoichiometric coefficient of the reacting species, in a balanced chemical equation.

Rate equation is always obtained from the experimental data.

1) $2NO_{(g)} + O_{2(g)} \rightarrow 2NO_{2(g)}$; Rate = $k[NO]^2[O_2]$

2) $CHCl_3 + Cl_2 \rightarrow CCl_4 + HCl$; Rate = $[CHCl_3][Cl_2]^{1/2}$.

3) $CH_3COOC_2H_5 + H_2O \rightarrow CH_3COOH + C_2H_5OH$; Rate = $k[CH_3COOC_2H_5]$

LAQ

185. Give two examples for each of zero order and first order reactions. Write the equations for the rate of a reaction in terms of concentration changes of reactants and products for the following reactions.

a) $A_{(g)} + B_{(g)} \rightarrow C_{(g)} + D_{(g)}$ **b)** $A_{(g)} \rightarrow B_{(g)} + C_{(g)}$ **c)** $A_{(g)} + B_{(g)} \rightarrow C_{(g)}$

A: **Two examples for zero order reaction:**

i) $2NH_{3(g)} \xrightarrow[1130K]{Pt} N_{2(g)} + 3H_{2(g)}$ ii) $2HI_{(g)} \xrightarrow[\Delta]{Au} H_{2(g)} + I_{2(g)}$

Two examples for first order reaction:

i) $NH_4NO_{2(g)} \rightarrow N_{2(g)} + 2H_2O$ ii) $SO_2Cl_{2(g)} \rightarrow SO_{2(g)} + Cl_{2(g)}$

For the reaction (a), $A_{(g)} + B_{(g)} \rightarrow C_{(g)} + D_{(g)}$; Rate = $-\frac{\Delta[A]}{\Delta t} = -\frac{\Delta[B]}{\Delta t} = +\frac{\Delta[C]}{\Delta t} = +\frac{\Delta[D]}{\Delta t}$

For the reaction (b), $A_{(g)} \rightarrow B_{(g)} + C_{(g)}$; Rate = $-\frac{\Delta[A]}{\Delta t} = +\frac{\Delta[B]}{\Delta t} = +\frac{\Delta[C]}{\Delta t}$

For the reaction (c), $A_{(g)} + B_{(g)} \rightarrow C_{(g)}$; Rate = $-\frac{\Delta[A]}{\Delta t} = -\frac{\Delta[B]}{\Delta t} = +\frac{\Delta[C]}{\Delta t}$

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186. Discuss the effect of temperature on the rate of a reaction. Derive necessary equations in this context.

A: Effect of temperature on rate of reaction: In general for every 10°C rise of temperature the rate of a reaction is nearly doubled and in some cases it is tripled.

The ratio of specific rates that differ by 10°C is known as temperature coefficient.

$$\frac{K_{t+10^\circ C}}{K_{t^\circ C}} = 2 \text{ (or) } 3$$

Effect of temperature on rate of reaction is explained by Arrhenius in the form of an equation known as Arrhenius Equation.

1) Arrhenius equation: The arrhenius equation gives accurately the temperature dependence of the rate of a chemical reaction.

2) Rate constant $k = A\, e^{-Ea/RT}$.....(i)

Where A is frequency factor, E_a is activation energy

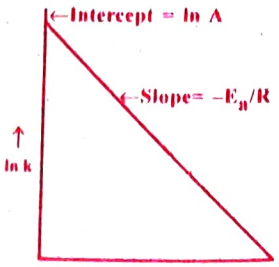
3) Taking natural logarithm of both sides of equation (i), we get $\ln k = \frac{-E_a}{RT} + \ln A$ .....(ii)

4) The plot of ln k vs 1/T is a straight line.

The slope of the line $= -\frac{E_a}{R}$

y-intercept = lnA.

Using these values, E_a and A can be calculated.



A plot between ln k and 1/T

5) At temperature T_1,

equation (ii) is $\ln k_1 = \frac{-E_a}{RT_1} + \ln A$ .....(iii)

6) At temperature T_2, equation (ii) is $\ln k_2 = \frac{-E_a}{RT_2} + \ln A$ .....(iv)

K_1 and K_2 are the values of rate constants at temperature T_1 and T_2 respectively.

7) Subtracting equation (iii) from (iv) we get, $\ln k_2 - \ln k_1 = \frac{E_a}{RT_1} - \frac{E_a}{RT_2}$ $[\because T_2 > T_1]$

8) $\ln \frac{k_2}{k_1} = \frac{E_a}{R}\left[\frac{1}{T_1} - \frac{1}{T_2}\right] \Rightarrow 2.303 \log \frac{k_2}{k_1} = \frac{E_a}{R}\left[\frac{1}{T_1} - \frac{1}{T_2}\right] \Rightarrow \log \frac{k_2}{k_1} = \frac{E_a}{2.303R}\left[\frac{1}{T_1} - \frac{1}{T_2}\right]$

d&f - BLOCK ELEMENTS (SAQ)

187. Explain why Ni and Fe exhibit zero oxidation state in $[Ni(CO)_4]$ and $[Fe(CO)_5]$ respectively.

A: In $[Ni(CO)_4]$ and $[Fe(CO)_5]$ the oxidation state of Ni and Fe is zero.
CO is a neutral ligand, hence the oxidation states of Ni and Fe are zero.
Low oxidation states are found when the complex compound has ligands capable of π-acceptor character in addition to the σ bonding.

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188. Write down the electronic configuration of (i) Cr^{3+} (ii) Cu^{+} (iii) Co^{2+} (iv) Mn^{2+}

A:
i) E.C of Cr^{3+} : $[Ar]3d^3$
ii) E.C of Cu^{+} : $[Ar]3d^{10}$
iii) E.C of Co^{2+} : $[Ar]3d^7$
iv) E.C of Mn^{2+} : $[Ar]3d^5$

$Cr(24) : Cr^{3+} (21 = 18 + 3)$
$Cu(29) : Cu^{+} (28 = 18 + 10)$
$Co(27) : Co^{2+} (25 = 18 + 7)$
$Mn(25) : Mn^{2+} (23 = 18 + 5)$

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189. Compare the stability of +2 oxidation state of the elements of the first transition series.

A: The sum $I.E_1 + I.E_2$ of first transition series increases due to increased nuclear charge.
Thus, the standard potentials become less and less negative.
So the tendency to form M^{+2} ion decreases.
The greater stability of +2 state of Mn is due to half-filled d-subshell (d^5).
For zinc, it is due to completely filled d-subshell (d^{10}).
For nickel, it is due to highest negative enthalpy of hydration.

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190. Compared to the changes in atomic and ionic sizes of elements of 3d and 4d series, the change in radii of elements of 4d and 5d series is virtually the same. Comment.

A: Compared to the change in atomic and ionic sizes of elements 3d and 4d series, the change in radii of elements 4d and 5d series is virtually the same.

This is due to the intervention of the 4f orbitals which must be filled before the 5d series of elements. This filling of 4f before 5d orbitals results in a regular decrease in atomic radii called Lanthanoid contraction. Due to Lanthanoid contraction the 4d and 5d series elements exhibit same size.

COORDINATION COMPOUNDS (SAQ)

191. What is meant by chelate effect ? Give example.

A: **Chelate effect:** It is the stabilization of complex compounds due to ring formation (chelation) by bidentate or polydentate ligands. Complexes containing chelating ligands are more stable than complexes containing unidentate ligands.

Ex: $[Co(en)_3]^{3+}$ is more stable than $[Co(NH_3)_6]^{3+}$

Note: en= Ethylenediamine $\Rightarrow$bidentate ligand (chelating ligand)

192. Give the oxidation numbers of the central metal atoms in the following complex entities
(i) $[Ni(CO)_4]$ (ii) $[Co(NH_3)_6]^{3+}$ (iii) $[Fe(CN)_6]^{4-}$ (iv) $[Fe(C_2O_4)_3]^{3-}$ [AP 19]

A: **i)** $[Ni(CO)_4]$

$x+4(0)=0$

$\Rightarrow x=0$

ii) $[Co(NH_3)_6]^{+3}$

$x+6(0) = +3$

$\Rightarrow x = +3$

In metal carbonyls metal oxidation state is equal to zero. (synergic effect)

iii) $[Fe(CN)_6]^{4-}$

$x+6(-1)= -4$

$\Rightarrow x=6-4=+2$

iv) $[Fe(C_2O_4)_3]^{3-}$

$x+3(-2) = -3$

$\Rightarrow x = +3$

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193. Explain the terms (i) Ligand (ii) Coordination number
(iii) Coordination entity (iv) Central metal atom/ ion

A: **i) Ligand:** The ions or molecules bound to the central atom / ion in the complex by donating lone pair of electrons are called ligands. a) simple ions such as Cl^-

b) small molecules such as H_2O

c) large molecules such as $H_2N–CH_2–CH_2–NH_2$

d) macro molecules such as proteins.

ii) Coordination Number (CN) : It is the number of ligands to which the metal is directly bonded. **Ex:** In $[Pt\ Cl_6]^{-2}$, C.N of Pt = 6

iii) Coordination entity: It is the central metal atom/ ion together with the ligands directly bonded to it. It is enclosed between square brackets. **Ex:** $[COCl_3(NH_3)_3]$, $[Ni(CO)_4]$

iv) Central metal atom /ion: The atom / ion to which a fixed number of ligands/ ions/ molecules are bound in a complex is called central metal atom or ion.

Ex: In $K_4[Fe(CN)_6]$ the central metal is Fe

HALOALKANES & HALOARENES (SAQ)

194. Write the mechanism of dehydrohalogenation of 2-bromobutane.

A: **Dehydrohalogenation of 2-bromobutane** is an **Elimination reaction.**

It involves the removal of one molecule of HX, from the molecule of a haloalkane.

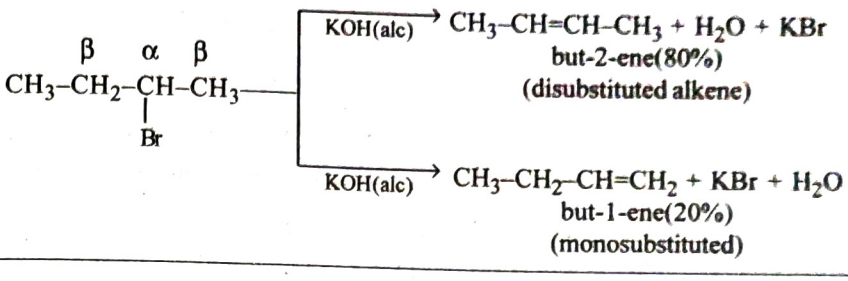
In elimination reactions, the halogen atom of the haloalkane (2-bromobutane) and a hydrogen atom from the adjacent carbon atom (β-carbon) are involved. Hence these reactions are also called β-elimination reactions. These reactions follow Saytzeff's rule.

Saytzeff rule: In dehydrohalogenation reactions, the preferred product is that alkene which has greater number of alkyl groups attached to the doubly bonded carbon atoms. i.e., more substituted alkane is the **major product** (but-2-ene(80%)).

Mechanism: In this reaction, but-2-ene is more substituted alkene than but-1-ene.

Hence, **but-2-ene is the major product.**

2-bromobutane can eliminate one molecule of HBr in the following two ways.



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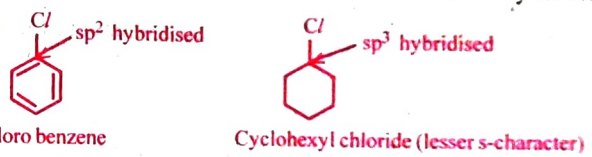
195. Explain why the dipole moment of chlorobenzene is lower than that of cyclohexyl chloride.

A: Carbon atom in chlorobenzene is sp^2 hybridised which is more electronegative than sp^3 carbon atom of cyclohexylchloride. Less electronegative leads to lesser s-character.
So, polarity of C–Cl bond in chlorobenzene is less than the C-Cl bond in cyclohexyl chloride.
Hence the dipole moment of chlorobenzene is lower than that of cyclohexyl chloride.



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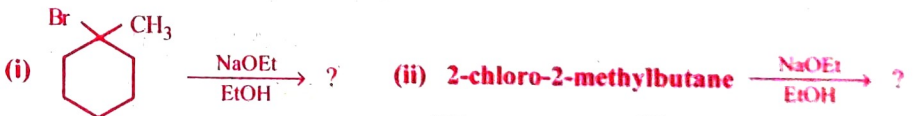
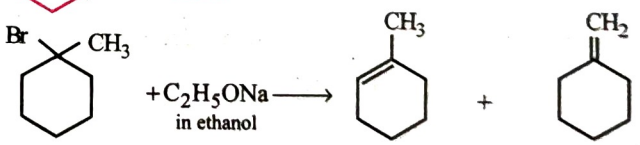
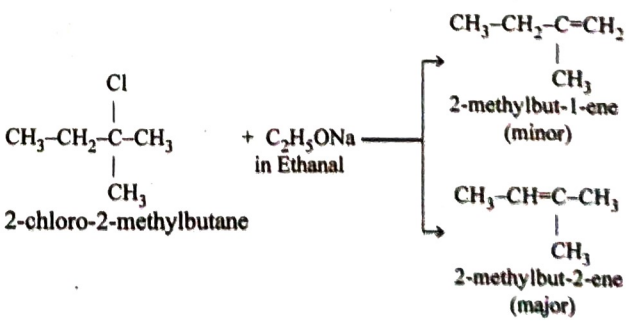
196. Predict the alkenes that would be formed in the following reactions and identify the major alkene.



A: (i)



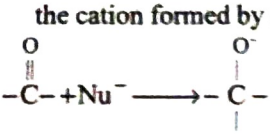
1-bromo 1-methyl cyclo hexane | 1-methyl cyclohex 1-ene (major) | Methylene cyclohexane (minor)



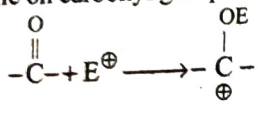
ALDEHYDES & KETONES (SAQ)

197. Explain why Aldehydes and ketones undergo nucleophilic addition while alkenes undergoes electrophilic addition though both are unsaturated compounds.

A: **1) Nucleophilic addition:** Aldehydes and ketones undergo nucleophilic addition because the intermediate anion formed by the attack of nucleophile on carbonyl group is more stable than the cation formed by attack of electrophile on carbonyl group.

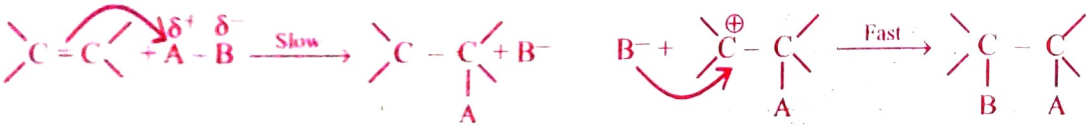


Anion(more stable)



Cation(less stable)

2) Electrophilic addition: Alkenes undergoes electrophilic addition beacuse in alkenes there is double bond between two carbon atoms and is electron rich. This double bond is highly reactive. So it is attacked by electrophilic first. So alkenes undergo electrophilic addition reactions.



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198. Arrange the following in the increasing order of their acidic strength: benzoic acid, 4-methoxybenzoic acid, 4-nitrobenzoic acid and 4-nethylbenzoic acid.

A: Acidic strength of electron with drawing $-NO_2$ group increases

Acidic strength of electron with releasing – $-OCH_3$, $-CH_3$ groups decreases

But releasing power of $-OCH_3$ is more than that of $-CH_3$.

Lower P^{ka} values indicate higher acidity.

Hence the increasing order (low to high) acidic strengths are given below:

4-methoxy benzoic acid < 4-methyl benzoic acid < benzoic acid < 4-nitrobenzoic acid.

P^{ka} values: (4.46) (4.36) (4.19) (3.41)

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CHAPTERS: 1,2,3

Q 1 UNIT TEST-1

Time : 1 Hour

SR. CHEMISTRY

Max.Marks : 25

Ans-Page Index

SECTION-A

I. Answer ALL the following questions. $3 \times 1 = 3$

1. Which law explains the solubility of gases in liquids ? [☞P 77(5)]
   1. Boyle's law 2. Charles' law 3.Henry's law 4.Raoult's law
2. Lead storage battery is an example of _____cell. [☞P 78(9)]
3. What is the term used for a solution that does not follow Raoult's law? [☞P 77(11)]

SECTION-B

II. Answer ALL the following questions $3 \times 2 = 6$

4. What are isotonic solutions? [☞P 56(80)]
5. State Faraday's first law of electrolysis. [☞P 58(86)]
6. What is standard hydrogen electrode? [☞P 59(94)]

SECTION-C

III. Answer ANY TWO of the following questions. $2 \times 4 = 8$

7. What is relative lowering of vapour pressure? How is it useful to determine the molar mass of a solute? [☞P 28(14)]
8. A solution of glucose in water is labeled as 10% w/w. What would be the molality of the solution ? [☞P 29(18)]
9. State and explain Nernst equation with the help of a metallic electrode and a non-metallic electrode. [☞P 32(23)]

(or)

10. What are Galvanic cells? Explain the working of Galvanic cell with a neat sketch taking Daniel cell as example. [☞P 30(19)]
11. State and explain Kohlrausch's law of independent migration of ions. [☞P 30(20)]
12. Define mass percentage, volume percentage and mass to volume percentage. [☞P 29(16)]

SECTION-D

IV. Answer ANY ONE of the following questions. $1 \times 8 = 8$

13. Give a detailed account of the collision theory of reaction rates of bimolecular gaseous reactions. [☞P 14(1)]

(or)

14. Explain the following terms with suitable examples. [☞P 16(3)]
    a) Average rate of a reaction b) Slow and fast reactions
    c) Order of a reaction d)Molecularity of a reaction
    e) Activation energy of a reaction

☞ **'Extra choice' is given in Section-C just to cover all the IMP questions**

CHAPTERS: 4,5,10

Q 2 UNIT TEST-2

SR. CHEMISTRY

Time : 1 Hour Max.Marks : 25

Ans-Page Index

SECTION-A

I. Answer ALL the following questions. $3 \times 1 = 3$

1. Lewis acid in the complex ion $[CoCl(NH_3)_5]^{2+}$ is [☞P 81(3)]

   1. Cl^- 2. NH_3 3. Co^{2+} 4. Co^{3+}

2. Mohr's salt formula is _____ [☞P 81(7)]

3. Name the disease that occurs due to deficiency of ascorbic acid. [☞P 86(14)]

SECTION-B

II. Answer ALL the following questions $3 \times 2 = 6$

4. Calculate the 'spin only' magnetic moment of $Fe^{2+}_{(aq)}$ ion. [☞P 62(109)]
5. $CuSO_4.5H_2O$ is blue in colour where as anhydrous $CuSO_4$ is colourless. Why? [☞P 65(122)]
6. What is Zwitter ion? Give an example. [☞P 74(170)]

SECTION-C

III. Answer ANY TWO of the following questions. $2 \times 4 = 8$

7. Write the characteristics properties of transition elements. [☞P 37(34)]
8. Explain Werner's theory of coordination compounds with suitable examples. [☞P 39(40)]
9. What are hormones? Give one example for each. [☞P 52(70)]
   (i) steroid hormones (ii) Polypeptide hormones and (iii) amino acid derivatives.

   (or)

10. What is lanthanoid contraction? What are the consequences of lanthanoid contraction? [☞P 37(35)]
11. Explain geometrical isomerism in coordination compounds giving suitable examples. [☞P 39(41)]
12. Write notes on vitamins. [☞P 53(71)]

SECTION-D

IV. Answer ANY ONE of the following questions. $1 \times 8 = 8$

13. a) Write four postulates of Werner's theory of coordination compounds. **(or)** [☞P 17(4)]
    b) Explain cis-trans isomerism in $[Pt(NH_3)_2Cl_2]$ complex.
14. Explain different types of isomerism exhibited by coordination compounds, giving suitable examples. [☞P 19(6)]

☞ **'Extra choice' is given in Section-C just to cover all the IMP questions**

CHAPTERS: 6,7,8,9

Q 3 UNIT TEST-3

Time : 1 Hour

SR. CHEMISTRY

Max.Marks : 25

SECTION-A

I. Answer ALL the following questions. $3 \times 1 = 3$

1. A primary alkyl halide would prefer to undergo
   1) S_N1 reaction 2) S_N2 reaction
   3) α–elimination 4) racemization
2. Alcohols and phenols react with carboxylic acids to form _____
3. The conversion of primary aromatic amines into diazonium salts is known as _____

SECTION-B

II. Answer ALL the following questions $3 \times 2 = 6$

4. What are ambident nucleophiles?
5. Identify the reactant needed to form t-butyl alcohol from acetone.
6. Write equations for carbylamine reaction of an aliphatic amine.

SECTION-C

III. Answer ANY TWO of the following questions. $2 \times 4 = 8$

7. Explain the mechanism of S_N1 with example.
8. Explain the following name reactions:
   i)Sandmeyer reaction ii) Gattermann reaction
9. Give the equations for the preparation of phenol from Cumene.

(or)

10. Write the products formed by the reduction and oxidation of phenol.
11. What is Tollen's reagent? Explain its reaction with Aldehydes.
12. Describe the following:
    i) Crossed aldol condensation ii) Decarboxylation

SECTION-D

IV. Answer ANY ONE of the following questions. $1 \times 8 = 8$

13. Explain the folliwng reactions
    a) Aldol reaction b) Esterification
    c) H.V.Z reaction d) Bromination of Benzoic acid
14. Describe the following:
    i) Acetylation ii) Cannizzaro reaction
    iii) Cross aldol condensation iv) Decarboxylation

☛ **'Extra choice' is given in Section-C just to cover all the IMP questions**

Ans-Page Index

[☛P 82(2)]
[☛P 83(8)]
[☛P 85(9)]
[☛P 66(129)]
[☛P 69(146)]
[☛P 72(156)]
[☛P 41(46)]
[☛P 49(64)]
[☛P 44(52)]
[☛P 44(53)]
[☛P 47(59)]
[☛P 48(63)]
[☛P 22(8)]
[☛P 23(9)]

Q 4 BOARD MODEL PAPER

Time : 3 Hours **SR. CHEMISTRY** Max.Marks : 85

SECTION-A

I. Answer ALL the following questions. $9 \times 1 = 9$

Ans-Page Index

1. Human blood is isotonic with _____solution (or) Saline. [☞P 77(10)]
2. Lead storage battery is an example of _____cell. [☞P 78(9)]
3. The number of collisions per second per unit volume of the reaction mixture is known as _____ [☞P 79(8)]
4. Which of the following coordination compound contains cobalt metal? [☞P 81(1)]
   1. Chlorophyll 2. Hemoglobin 3. Vitamin B_{12} 4. Myoglobin
5. Which of the following is the suitable medium for preparing Grignard reagents? [☞P 82(3)]
   1)Dry acetone 2) Dry ether 3)Concentrated HCl 4)Alcoholic KOH
6. Glycerol is a: [☞P 83(1)]
   1)dihydric alcohol 2) monohydric alcohol
   3) hexahydric alcohol 4) trihydric alcohol
7. What is Rochelle salt? [☞P 84(12)]
8. What is the IUPAC name of ethyl methyl amine? [☞P 85(11)]
9. Which linkage connects two monosaccharide units? [☞P 86(11)]

SECTION-B

II. Answer ALL the following questions $14 \times 2 = 28$

10. State Henry's law. [☞P 56(78)]
11. Define osmotic pressure. [☞P 56(79)]
12. State Faraday's first law of electrolysis. [☞P 58(86)]
13. What is metallic corrosion? Give one example. [☞P 58(89)]
14. What is a pseudo first order reaction? Give one example. [☞P 60(96)]
15. Why Zn^{2+} is diamagnetic whereas Mn^{2+} is paramagnetic? [☞P 62(108)]
16. Calculate the 'spin only' magnetic moment of $Fe^{2+}_{(aq)}$ ion. [☞P 62(109)]
17. What is an ambidentate ligand? Give example. [☞P 64(121)]
18. What are Enantiomers? [☞P 66(127)]
19. Give the reagents used for the preparation of phenol from chlorobenzene. [☞P 68(137)]
20. List the reagents suitable to reduce carboxylic acid to alcohol. [☞P 70(147)]
21. Arrange the following bases in increasing order of their basic strength. Aniline, p-nitroaniline and p-toluidine. [☞P 72(155)]
22. Differentiate between globular and fibrous proteins. [☞P 74(167)]
23. What are essential and non-essential amino acids? Give one example for each. [☞P 74(165)]

SECTION-C

III. Answer ANY EIGHT of the following questions. $8 \times 4 = 32$

24. What is meant by positive deviations from Raoult's law and how is the sign of Δ_{mix} H related to positive deviation from Raoult's law ? |☞P 27(12)|
25. State and explain Kohlrausch's law of independent migration of ions. |☞P 30(20)|
26. What is 'molecularity' of a reaction? How is it different from the 'order' of a reaction? Name one bimolecular and one trimolecular gaseous reactions. |☞P 34(28)|
27. What are Galvanic cells? Explain the working of Galvanic cell with a neat sketch taking Daniel cell as example. |☞P 30(19)|
28. Write the characteristics properties of transition elements. |☞P 37(34)|
29. What are homoleptic and heteroleptic complexes? Give one example for each. |☞P 40(42)|
30. Explain the mechanism of S_N1 with example. |☞P 41(46)|
31. Explain the acidic nature of phenols and compare with that of alcohols. |☞P 44(54)|
32. Give the equations for the preparation of phenol from Cumene. |☞P 44(52)|
33. What is Tollen's reagent? Explain its reaction with Aldehydes. |☞P 47(59)|
34. Explain the following reactions: |☞P 49(64)|
    i)Sandmeyer reaction ii) Gattermann reaction
35. Give the dietary sources of the following vitamins and name the diseases caused by their deficiency (a) A (b) D (c) E and (d) K |☞P 52(69)|

SECTION-D

IV. Answer ANY TWO of the following questions. $2 \times 8 = 16$

36. Give a detailed account of the collision theory of reaction rates of bimolecular gaseous reactions. |☞P 14(1)|
37. a)Write four postulates of Werner's theory of coordination compounds. |☞P 17(4)|
    b)Explain cis-trans isomerism in $[Pt(NH_3)_2Cl_2]$ complex.
37. Explain the folliwng reactions |☞P 22(8)|
    a) Aldol reaction b) Esterification
    c) H.V.Z reaction d) Bromination of Benzoic acid

GUESS PAPERS

Q 5 GUESS PAPER -1

SR. CHEMISTRY

Time : 3 Hours | Max.Marks : 85

SECTION-A

Ans-Page Index

I. Answer ALL the following questions. $9 \times 1 = 9$

1. Solubility of a gas in liquid _____ with increase in pressure. [☞P 77(7)]
2. 1 Faraday value is equal to _____ C mol^{-1} [☞P 78(6)]
3. Which reactions takes place in one step? [☞P 79(12)]
4. Lewis acid in the complex ion $[CoCl(NH_3)_5]^{2+}$ is [☞P 81(3)]
   1. Cl^- 2. NH_3 3. Co^{2+} 4. Co^{3+}
5. What is the general formula for haloalkanes? (X=halogen atom, n=1, 2, 3…) [☞P 82(1)]
   1) $C_nH_{2n}X$ 2) $C_nH_{2n+1}X$ 3) $C_nH_{2n-1}X$ 4) $C_nH_{2n-3}X$
6. Which of the following shows strong hydrogen bonding? [☞P 83(3)]
   1) Ethyl amine 2)Ammonia 3)Ethyl alcohol 4)Diethyl ether
7. Which acid is used as solvent in food industry as vinegar? [☞P 84(13)]
8. Coupling reaction of aryl diazonium salts with phenols or aryl amines gives rise to the formation of_____ [☞P 85(10)]
9. Which non-essential amino acid is optically inactive? [☞P 86(12)]

SECTION-B

II. Answer ALL the following questions $14 \times 2 = 28$

10. What are isotonic solutions? [☞P 56(80)]
11. Calculate the molarity of a solution containing 5 g of NaOH in 450 ml solution. [☞P 57(84)]
12. State Faraday's second law of electrolysis. [☞P 58(87)]
13. What is cell constant of a conductivity cell? [☞P 59(92)]
14. Give two examples for gaseous first order reactions. [☞P 60(97)]
15. Aqueous Cu^{2+} ions are blue in colour, where as Aqueous Zn^{2+} ions are colourless. Why? [☞P 62(110)]
16. What is mischmetal? Give its composition and uses. [☞P 63(112)]
17. $CuSO_4.5H_2O$ is blue in colour where as anhydrous $CuSO_4$ is colourless. Why? [☞P 65(122)]
18. What are ambident nucleophiles? [☞P 66(129)]
19. Explain why propanol has higher boiling point than that of the hydrocarbon butane. [☞P 68(138)]
20. Write the reaction showing α-halogenation of carboxylic acid and give its name. [☞P 70(148)]
21. Write equations for carbylamine reaction of an aliphatic amine. [☞P 72(156)]
22. What are anomers? [☞P 74(169)]
23. Why are sugars classified as reducing and non-reducing sugars? [☞P 75(172)]

SECTION-C

III. Answer ANY EIGHT of the following questions. **8 × 4 = 32**

24. What is relative lowering of vapour pressure? How is it useful to determine the molar mass of a solute? [☞P 28(14)]
25. Give the construction and working of a standard hydrogen electrode with a neat diagram. [☞P 31(21)]
26. Derive the integrated rate equation for a zero order reaction. [☞P 35(30)]
27. State and explain Kohlrausch's law of independent migration of ions. [☞P 30(20)]
28. What is lanthanoid contraction? What are the consequences of lanthanoid contraction? [☞P 37(35)]
29. Explain Werner's theory of coordination compounds with suitable examples. [☞P 39(40)]
30. How will you carry out the following conversions? [☞P 41(47)]
    (i) Ethane to bromoethane (ii) Toluene to benzyl alcohol
31. Write the products formed by the reduction and oxidation of phenol. [☞P 44(53)]
32. Account for the statement: Alcohols boil at highest temperature than hydrocarbons and ethers of comparable molecular masses. [☞P 45(56)]
33. Write the oxidation products of acetaldehyde, acetone and acetophenone. [☞P 47(60)]
34. Write the reactions of [☞P 50(65)]
    (i) aromatic and (ii) aliphatic primary amines with nitrous acid, highlighting the differences.
35. What are hormones? Give one example for each. [☞P 52(70)]
    (i) steroid hormones (ii) Polypeptide hormones and (iii) amino acid derivatives.

SECTION-D

IV. Answer ANY TWO of the following questions. **2 × 8 = 16**

36. a) Write the differences between order & molecularity. [☞P 15(2)]
    b) Describe the salient features of collision theory.
37 Discuss the nature of bonding and magnetic behaviour in the following coordination entities on the basis of valence bond theory [☞P 20(7.1)]
    (i) $[Fe(CN)_6]^{4-}$ (ii) $[FeF_6]^{3-}$
38. Describe the following: [☞P 23(9)]
    i) Acetylation ii) Cannizzaro reaction
    iii) Cross aldol condensation iv) Decarboxylation

GUESS PAPERS

Q 6 GUESS PAPER -2

Time : 3 Hours

SR. CHEMISTRY

Max.Marks : 85

SECTION-A

I. Answer ALL the following questions. $9 \times 1 = 9$

1. Desalination plants work on the basis of the phenomenon called _____ [☞P 77(9)]
2. In Daniell cell the oxidation half reaction occurs at _____ [☞P 78(10)]
3. What is the unit of rate constant for a zero-order reaction? [☞P 79(14)]
4. Which ligand gives a stable complex? [☞P 81(4)]
   1. SCN^- 2. Cl^- 3. H_2O 4. Ethylene diamine(en)
5. A primary alkyl halide would prefer to undergo [☞P 82(2)]
   1) S_N1 reaction 2) S_N2 reaction 3) α–elimination 4) racemization
6. Which of the following can work as a dehydrating agent for alcohols? [☞P 83(2)]
   1) Conc. H_2SO_4 2) Anhydrous Al_2O_3 3) H_3PO_4 4) All of these
7. Aldehydes and ketones undergo what type of addition reactions? [☞P 84(14)]
8. A molecule that has both positive and negative regions of charge is called [☞P 85(15)]
9. Name the disease that occurs due to deficiency of ascorbic acid. [☞P 86(14)]

SECTION-B

II. Answer ALL the following questions $14 \times 2 = 28$

10. What is Cryoscopic constant? [☞P 57(82)]
11. Define mole fraction. [☞P 56(77)]
12. What is a fuel cell? How is it different from a conventional galvanic cell? [☞P 58(88)]
13. Define molar conductivity Λ_m and how is it related to conductivity (k)? [☞P 59(91)]
14. Give two examples for zero Order reactions. [☞P 60(98)]
15. What is an alloy? Give example. [☞P 62(111)]
16. What is actinoid contraction? [☞P 63(113)]
17. What is a double salt? Give example. [☞P 64(118)]
18. What type of isomerism is exhibited by o-, m- and p-dichlorobenzenes? [☞P 66(128)]
19. Alcohols are comparatively more soluble in water than hydrocarbons of comparable molecular masses. Explain this fact. [☞P 68(139)]
20. Compare the acidic strength of acetic acid, Chloroacetic acid, benzoic acid and Phenol. [☞P 71(153)]
21. Gabriel phthalimide synthesis exclusively forms primary amines . Explain. [☞P 72(157)]
22. Name the vitamin responsible for the coagulation of blood. [☞P 74(166)]
23. What is Zwitter ion? Give an example. [☞P 74(170)]

Ans-Page Index

SECTION-C

III. Answer ANY EIGHT of the following questions. $8 \times 4 = 32$

24. What is meant by negative deviation from Raoult's law and how is the sign of Δ_{mix} H related to negative deviation from Raoult's law ? [☞P 27(13)]
25. What are the products obtained at the cathode and the anode during the electrolysis of the following when platinum electrodes are used in the electrolysis(a) Molten KCl(b) Aq. $CuSO_4$ solution (c) Aq.K_2SO_4 solution [☞P 31(22)]
26. What is half-life ($t_{1/2}$) of a reaction? Derive the equations for the 'half-life' value of zero and first order reactions. [☞P 35(31)]
27. What are Galvanic cells? Explain the working of Galvanic cell with a neat sketch taking Daniel cell as example. [☞P 30(19)]
28. What are interstitial compounds? How are they formed? Give two examples. [☞P 37(36)]
29. Explain geometrical isomerism in coordination compounds giving suitable examples. [☞P 39(41)]
30. Describe the stereochemical effect on the hydrolysis of 2-bromobutane. [☞P 42(48)]
31. Write the mechanism of hydration of ethene to yield ethanol. [☞P 45(57)]
32. While separating a mixture of ortho and para nitrophenols by steam distillation, name the isomer which will be steam volatile. Give reason. [☞P 45(55)]
33. Write the equations of any aldehyde with Fehling's reagent. [☞P 47(61)]
34. Write the equations involved in the reaction of Nitrous acid with ethylamine and aniline. [☞P 50(66)]
35. Explain the denaturation of proteins. [☞P 54(73)]

SECTION-D

IV. Answer ANY TWO of the following questions. $2 \times 8 = 16$

36. Explain the following terms with suitable examples. [☞P 16(3)]
    a) Average rate of a reaction
    b) Slow and fast reactions
    c) Order of a reaction
    d)Molecularity of a reaction
    e) Activation energy of a reaction
37. Explain different types of isomerism exhibited by coordination compounds, giving suitable examples. [☞P 19(6)]
38. Explain the following terms. Give an example of the reaction in each case. [☞P 25(11)]
    i) Cyanohydrin
    ii) Acetal
    iii) Semicarbazone
    iv)Aldol

Q 7

GUESS PAPERS

GUESS PAPER -3

Time : 3 Hours

SR. CHEMISTRY

Max.Marks : 85

Ans-Page Index

SECTION-A

I. Answer ALL the following questions. $9 \times 1 = 9$

1. For an ideal solution, $\Delta_{mix}H$ is equal to _____ [☞P 77(8)]
2. What is the charge of an electron in coulombs? [☞P 78(13)]
3. The sum of powers of the concentration terms in the rate law expression is called _____ of that chemical reaction. [☞P 79(6)]
4. Which of the following is not a double salt? [☞P 81(2)]
   1) $FeSO_4.(NH_4)_2SO_4.6H_2O$ 2) $KCl.MgCl_2 6H_2O$
   3) $PtCl_4.2HCl$ 4) $KAl(SO_4)_2.12H_2O$
5. Which of the following alkyl halide will undergo S_N1 reaction most readily? [☞P 82(4)]
   1) $(CH_3)_3C—F$ 2) $(CH_3)_3C—Cl$ 3) $(CH_3)_3C—Br$ 4) $(CH_3)_3C—I$
6. When phenol is treated with excess bromine water it gives [☞P 83(4)]
   1) m-bromophenol 2) o- and p-bromophenol
   3) 2,4-dibromophenol 4) 2,4,6-tribromophenol
7. What is name of the product formed when benzene is reacted with CO and HCl in the presence of anhydrous $AlCl_3$? [☞P 84(11)]
8. Aniline and other arylamines are usually colourless but get coloured on storage why? [☞P 85(13)]
9. Increased level of which hormone causes hyperthyroidism ? [☞P 86(15)]

SECTION-B

II. Answer ALL the following questions $14 \times 2 = 28$

10. Define molarity. [☞P 56(75)]
11. Calculate the amount of benzoic acid (C_6H_5COOH) required for preparing 250 mL of 0.15 M solution of benzoic acid in methanol. [☞P 57(83)]
12. State Faraday's first law of electrolysis. [☞P 58(86)]
13. What is standard hydrogen electrode? [☞P 59(94)]
14. Write the equation for the rate of the reaction [☞P 60(100)]

$$5Br^-_{(aq)} + BrO^-_{3(aq)} + 6H^+_{(aq)} \rightarrow 3Br_{2(aq)} + 3H_2O(1)$$

15. What are transition elements? Give examples. [☞P 62(107)]
16. Calculate the 'spin only' magnetic moment of $Fe^{2+}_{(aq)}$ ion. [☞P 62(109)]
17. What are coordination compounds? Give two examples. [☞P 64(117)]
18. Write the isomers of the compound having molecule formula C_4H_9Br? [☞P 66(132)]
19. Preparation of ethers by acid dehydration of secondary or tertiary alcohols is not a suitable method. Give reason. [☞P 68(141)]

20. How do you distinguish acetophenone and benzophenone? [☛P 70(150)]
21. Explain why ethylamine is more soluble in water where as aniline is sparingly soluble. [☛P 73(161)]
22. Write the names of three types of RNA. [☛P 75(171)]
23. What is glycogen? How is it different from starch? [☛P 74(168)]

SECTION-C

III. Answer ANY EIGHT of the following questions. $8 \times 4 = 32$

24. Calculate the mass of a non-volatile solute (molar mass 40g mol^{-1}) which should be dissolved in 114g Octane to reduce its vapour pressure to 80%. [☛P 28(15)]
25. State and explain Nernst equation with the help of a metallic electrode and a non-metallic electrode. [☛P 32(23)]
26. Derive an integrated rate equation for a first order reaction. [☛P 35(30)]
27. Give the construction and working of a standard hydrogen electrode with a neat diagram. [☛P 31(21)]
28. Write the characteristics of interstitial compounds. [☛P 38(37)]
29. Using IUPAC norms write the formulas for the following : [☛P 40(44)]
    (i) Tetrahydroxozincate (II) ion (ii) Hexaaminecobalt (III) sulphate
    (iii) Potassium tetrachloropalladate(II)
    (iv) Potassium tri(oxoalato)chromate(III)
30. Write the structures of the following organic halides. [☛P 43(51)]
    (i) 1-bromo-4-sec-butyl-2 methylbenzene (ii)2-chloro-1-phenylbutane
    (iii) p-bromochlorobenzene (iv)4-t-butyl-3-iodoheptane
31. Draw the structures of all isomeric alcohols of molecular formula $C_5H_{12}O$ and give their IUPAC names and classify them as primary, secondary, and tertiary alcohols. [☛P 46(58)]
32. Write the products formed by the reduction and oxidation of phenol. [☛P 44(53)]
33. Write the IUPAC names of the following [☛P 48(62)]
    (i) $CH_3CH_2CH(Br)CH_2COOH$ (ii) $PhCH_2COCH_2COOH$
    (iii) $CH_3CH(CH_3)CH_2COOC_2H_5$
34. Compare the basicity of the following in gaseous state and in aqueous state and arrange them in increasing order of basicity. [☛P 51(68)]
    CH_3NH_2, $(CH_3)_2NH$, $(CH_3)_3N$ and NH_3
35. Write notes on vitamins. [☛P 53(71)]

SECTION-D

IV. Answer ANY TWO of the following questions. $2 \times 8 = 16$

36. Give a detailed account of the collision theory of reaction rates of bimolecular gaseous reactions. [☛P 14(1)]
37 Discuss the nature of bonding and magnetic behaviour in the following coordination entities on the basis of valence bond theory [☛P 21(7.2)]
    (i) $[Co(C_2O_4)_3]^{3-}$ (ii) $[CoF_6]^{3-}$
38. How do you distinguish the following pairs of compounds? [☛P 24(10)]
    i) Propanal and propanone ii) Acetophenone and benzophenone
    iii) Phenol and benzoic acid iv) Pentan-2-one and Pentan-3-one

Q 8

GUESS PAPERS

GUESS PAPER -4

SR. CHEMISTRY

Time : 3 Hours | Max.Marks : 85

	Ans-Page Index

SECTION-A

I. Answer ALL the following questions. $9 \times 1 = 9$

1. The units of boiling point elevation constant K_b are _____ [☞P 77(6)]
2. Corrosion of iron is mainly due to _____ and _____ [☞P 78(7)]
3. The overall rate of the reaction is controlled by the slowest step of the reaction and is called the _____ [☞P 79(7)]
4. Which complex shows geometrical, optical and also structural isomerism? [☞P 81(5)]
   1. $[Co(en)_3]Cl_3$ 2. $[CoCl_2(en)_2]Br$ 3. $[Co(en)(NH_3)_2Cl_2]Cl$ 4. $[PdCl_2(NH_3)_2$
5. Chlorobenzene is [☞P 82(5)]
   1) less reactive than benzyl chloride 2) more reactive than ethyl bromide
   3) nearly as reactive as methyl chloride 4) more reactive than isopropyl chloride
6. Which of the following is formed when phenol is exposed to air? [☞P 83(5)]
   1) Benzoquinone 2) Benzyl quinone 3) Acetophenone 4) Benzene
7. Phthalic acid on reaction with ammonia followed by strong heating gives a compound 'X' along with elimination of NH_3. What is the name of that compound 'X'? [☞P 84(15)]
8. The process of cleavage of the C–X bond of alkyl halides by ammonia molecule is known as [☞P 85(12)]
9. What type of bonding helps in stabilizing the α-helix structure of proteins? [☞P 86(13)]

SECTION-B

II. Answer ALL the following questions $14 \times 2 = 28$

10. Define molality. [☞P 56(76)]
11. What is Ebullioscopic constant? [☞P 57(81)]
12. Give the electro-chemical reaction that represents the corrosion or rusting of iron. [☞P 58(90)]
13. How is a galvanic cell represented on paper as per IUPAC convention? Give one example. [☞P 59(93)]
14. What are complex reactions? Name one complex reaction. [☞P 61(103)]
15. What is mischmetal? Give its composition and uses. [☞P 63(112)]
16. Why Zn^{2+} is diamagnetic whereas Mn^{2+} is paramagnetic? [☞P 62(108)]
17. What is the oxidation state of cobalt in (i) $K[Co(CO)_4]$ and (ii) $[Co(NH_3)_6]^{3+}$ [☞P 65(123)]
18. Write the structure of the following compounds [☞P 66(131)]
    (i) 2-chloro-3-methylpentane (ii) 1-bromo-4-sec-butyl-2-methylbenzene
19. Write the structures for the following compounds [☞P 69(143)]
    1) Ethoxy Ethane 2) Ethoxybutane 3) Phenoxy ethane
20. An organic acid with molecular formula $C_8H_8O_2$ on decarboxylation forms Toluene. Identify the organic acid. [☞P 70(151)]

21. Arrange the following bases in decreasing order of pK_b values $C_2H_5NH_2$, $C_6H_5NHCH_3$, $(C_6H_5)_2NH$ and $C_6H_5NH_2$. [☛P 72(158)]
22. What is Zwitter ion? Give an example. [☛P 74(170)]
23. What is the difference between a nucleoside and a nucleotide? [☛P 75(174)]

SECTION-C

III. Answer ANY EIGHT of the following questions. $8 \times 4 = 32$

24. A solution of glucose in water is labeled as 10% w/w. What would be the molality of the solution ? [☛P 29(18)]
25. What is metallic corrosion? Explain it with respect to iron corrosion. [☛P 32(24)]
26. What is Arrhenius equation? Derive an equation which describes the effect of rise of temperature (T) on the rate constant (k) of a reaction. [☛P 36(33)]
27. What are Galvanic cells? Explain the working of Galvanic cell with a neat sketch taking Daniel cell as example. [☛P 30(19)]
28. Why do the transition metal ions exhibit characteristic colours in aqueous solution. Explain giving examples. [☛P 38(38)]
29. What are homoleptic and heteroleptic complexes? Give one example for each. [☛P 40(42)]
30. Write the differences between S_N1 and S_N2 reactions. [☛P 42(49)]
31. Account for the statement: Alcohols boil at highest temperature than hydrocarbons and ethers of comparable molecular masses. [☛P 45(56)]
32. Write the mechanism of hydration of ethene to yield ethanol. [☛P 45(57)]
33. Describe the following: [☛P 48(63)]
    i) Crossed aldol condensation ii) Decarboxylation
34. Give one chemical test to distinguish between the following pairs of compounds: [☛P 50(67)]
    i) Methylamine and dimethylamine ii) Aniline and N-methylaniline
    iii) Ethylamine and aniline
35. What are enzymes?Give examples. [☛P 54(74)]

SECTION-D

IV. Answer ANY TWO of the following questions. $2 \times 8 = 16$

36. a)Write the differences between order & molecularity. [☛P 15(2)]
    b)Describe the salient features of collision theory.
37. a)Write four postulates of Werner's theory of coordination compounds. [☛P 17(4)]
    b) Explain cis-trans isomerism in $[Pt(NH_3)_2Cl_2]$ complex.
38. Explain the folliwng reactions [☛P 22(8)]
    a) Aldol reaction b) Esterification
    c) H.V.Z reaction d) Bromination of Benzoic acid

Q 9

GUESS PAPERS

GUESS PAPER -5

SR. CHEMISTRY

Time : 3 Hours Max.Marks : 85

Ans-Page Index

SECTION-A

I. Answer ALL the following questions. 9 × 1 = 9

1. What is the name given to the solutions, when they have same osmotic pressure at a given temperature? [☛P 77(15)]
2. The units of molar conductivity are _____ [☛P 78(8)]
3. $k = A\ e^{-Ea/RT}$ where A is called _____ [☛P 79(9)]
4. The IUPAC name of $K_2[PdCl_4]$ is _____ [☛P 81(9)]
5. A primary alkyl halide would prefer to undergo [☛P 82(2)]
   1) S_N1 reaction 2) S_N2 reaction
   3) α–elimination 4) racemization
6. When phenol is treated with excess bromine water it gives [☛P 83(4)]
   1) m-bromophenol 2) o- and p-bromophenol
   3) 2,4-dibromophenol 4) 2,4,6-tribromophenol
7. Which of the following is a dicarboxylic acid? [☛P 84(3)]
   1) oxalic acid 2) benzoic acid 3) salicylic acid 4) picric acid
8. The process of cleavage of the C–X bond of alkyl halides by ammonia molecule is known as [☛P 85(12)]
9. Name the disease that occurs due to deficiency of ascorbic acid. [☛P 86(14)]

SECTION-B

II. Answer ALL the following questions 14 × 2 = 28

10. Define mole fraction. [☛P 56(77)]
11. Calculate molality of 2.5 g of ethanoic acid (CH_3COOH) in 75g of benzene. [☛P 57(85)]
12. State Faraday's first law of electrolysis. [☛P 58(86)]
13. Write the Nernst equation for the EMF of the cell [☛P 59(95)]
    $Ni(s) / Ni^{2+}(aq) // Ag^{+}(aq) / Ag(s)$
14. A first order reaction is found to have a rate constant $k = 5.5 \times 10^{-14}\ s^{-1}$. Find the half-life of the reaction. [☛P 61(106)]
15. How do transition metals exhibit catalytic activity? [☛P 63(115)]
16. Why do the transition metals form a large number of complex compounds? [☛P 63(114)]
17. $[Cr(NH_3)_6]^{3+}$ is paramagnetic while $[Ni(CN)_4]^{2-}$ is diamagnetic. Why? [☛P 65(124)]
18. What is the stereochemical outcome of S_N1 and S_N2 reactions? [☛P 66(130)]
19. Identify the reactant needed to form t-butyl alcohol from acetone. [☛P 69(146)]
20. Write equations showing the conversion of [☛P 71(154)]
    i) acetic acid to acetyl chloride ii) benzoic acid to benzamide
21. Give structures of A,B and C in the following reaction. [☛P 72(160)]

$$C_6H_5N_2Cl \xrightarrow{CuCN} A \xrightarrow{H_2O/H^+} B \xrightarrow{NH_3} C$$

22. Write the names of three types of RNA. [☛P 75(171)]
23. Name the different types of carbohydrates on the basis of their hydrolysis. Give one example for each. [☛P 75(173)]

SECTION-C

III. Answer ANY EIGHT of the following questions. $8 \times 4 = 32$

24. Define mass percentage, volume percentage and mass to volume percentage solutions. [☛P 29(16)]
25. What are fuel cells? How are they different from galvanic cells? Give the construction of H_2-O_2 fuel cell. [☛P 88(179)]
26. Explain the terms [☛P 36(32)]
    a) Activation energy (E_a) b) Collision frequency (Z)
    c) Probability factor (P) with respect to Arrhenius equation.
27. What is lanthanoid contraction? What are the consequences of lanthanoid contraction? [☛P 37(35)]
28. A solution of $CuSO_4$ is electrolysed for 10 minutes with a current of 1.5 amperes. What is the mass of copper deposited at the cathode? [☛P 33(25)]
29. Explain the following terms by giving an example [☛P 40(43)]
    i) Unidentate ligand ii)Bidentate ligand
    iii) Polydentate ligand iv)Ambidentate ligand
30. Give the IUPAC names of the following compounds. [☛P 43(50)]
    (i) CH_3–CH(Cl)–CH(I)CH_3 (ii) $ClCH_2$.CH=CH–CH_2Br
    (iii) $(CCl_3)_3CCl$ (iv) $CH_3C(p\text{–}Cl\text{–}C_6H_4)_2$–CH(Br)$CH_3$
31. Give the equations for the preparation of phenol from Cumene. [☛P 44(52)]
32. Draw the structures of all isomeric alcohols of molecular formula $C_5H_{12}O$ and give their IUPAC names and classify them as primary, secondary, and tertiary alcohols. [☛P 46(58)]
33. Arrange the following in the increasing order of their acidic strength: benzoic acid, 4-methoxybenzoic acid, 4-nitrobenzoic acid and 4-nethylbenzoic acid. [☛P 96(198)]
34. Write the equations involved in the reaction of Nitrous acid with ethylamine and aniline. [☛P 50(66)]
35. Write the importance of carbohydrates. [☛P 54(72)]

SECTION-D

IV. Answer ANY TWO of the following questions. $2 \times 8 = 16$

36. Give a detailed account of the collision theory of reaction rates of bimolecular gaseous reactions. [☛P 14(1)]
37. Explain the applications of coordination compounds in different fields. [☛P 18(5)]
38. Describe the following: [☛P 23(9)]
    i) Acetylation ii) Cannizzaro reaction
    iii) Cross aldol condensation iv) Decarboxylation

QR CODE SCANNER PAGE

1. SOLUTIONS

2. ELECTRO CHEMISTRY

3. CHEMICAL KINETICS

4. THE d&f BLOCK ELEMENTS

5. COORDINATION COMPOUNDS

6. HALOALKANES & HALOARENES

7. ALCOHOLS, PHENOLS & ETHERS

8. ALDEHYDES, KETONES & CARBOXYLIC ACIDS

9. AMINES

10. BIOMOLECULES

IMP FORMULAS

CHEM BEATS

TOP 25 NAMED REACTIONS

OMQ – Hints & Solutions

BBQ
BEYOND MARKS

BBQ HEROES

TOP STORIES THAT SAY: "I CAN DO IT!"

Your attitude today builds your greatness tomorrow!

KALPANA CHAWLA

Dreams Take You Far

From Karnal to space, she pursued her dream and became the first woman of Indian origin to fly in space.

LESSON: Dream beyond your surroundings.

YOUR TURN: Write down one big dream of yours.

P. V. SINDHU

Keep Fighting

She faced many defeats and setbacks but never stopped. She became an Olympic medallist and World Champion.

LESSON: Champions don't fear another attempt.

YOUR TURN: Don't quit. Try once more.

D. GUKESH

Youngest World Champion

At just 18, he became the youngest-ever undisputed World Chess Champion.

LESSON: Age is not a limit to achievement.

YOUR TURN: Believe in yourself. Great things are possible!

KONERU HUMPY

Patience Wins

A chess prodigy at a young age, she continued to compete at the highest level and inspired millions.

LESSON: Talent grows through patience and persistence.

YOUR TURN: Be patient. Keep learning every day.

ARUNIMA SINHA

Nothing Is Impossible

After losing a leg in an accident, she set a goal and became the first Indian amputee to climb Mount Everest.

LESSON: Don't let one setback define your life.

YOUR TURN: Decide your goal and never give up.

MITHALI RAJ

Let Performance Speak

She faced the challenges of being a woman in cricket and became one of India's greatest cricketers and a long-serving captain.

LESSON: Stay disciplined. Let your performance speak.

YOUR TURN: Do your best in everything you do.

SUDHA CHANDRAN

Dance of Courage

After losing a leg, she learned to dance again with a prosthetic limb and returned to the stage and screen.

LESSON: What you lose need not take away your dream.

YOUR TURN: Face your challenges with courage.

SACHIN TENDULKAR

Practice. Patience. Passion.

Started young, faced failures and injuries, but built his greatness through years of practice, patience and discipline.

LESSON: Greatness is built one day at a time.

YOUR TURN: Be consistent every single day.

SAINA NEHWAL

Fall. Recover. Return Stronger.

She faced injuries and tough phases but worked hard and came back to reach the top of world badminton.

LESSON: Fall. Recover. Return stronger.

YOUR TURN: Come back stronger from every fall.

NAMBI NARAYANAN

Truth. Patience. Purpose.

A brilliant space scientist, he faced a false case and years of hardship. He never gave up and continued to serve the nation.

LESSON: Truth may be delayed, but never stop standing for it.

YOUR TURN: Always choose honesty and hard work.

GOODNESS ACTIONS (NO MONEY NEEDED!)

- Plant a Sapling
- Make Someone Smile
- Read & Share Good Books
- Save Water
- Help Someone
- Keep It Clean
- Listen Patiently
- Encourage Others
- Give Without Expecting
- Respect Your Nation

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